

Climate Chain: From Gas to Warming

I can explain how extra greenhouse gases raise Earth's average temperature and connect warming to wider climate effects.

**Weather is what happens now
climate is a pattern over many years**

**Extra greenhouse gases mean
heat energy leaves Earth more slowly**

Use a cause chain to explain an effect without blaming individuals

[Open the original interactive lesson](#)

What success looks like

- I can distinguish weather from climate.
- I can order the enhanced-greenhouse causal chain.
- I can connect warming to one likely effect and name a model limitation.

Retrieve and reconnect

- State one difference between weather and climate.
- Recall why repeated measurements are useful.
- Name one source of energy that warms Earth's surface.
- Explain why one cold day cannot describe a long-term pattern.

It is cold today. Which evidence could tell us about climate?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Today's temperature at one place
- B. A long record of averages from many places
- C. One photograph of snow

Look closely · what is the evidence?

WEATHER · what happens now

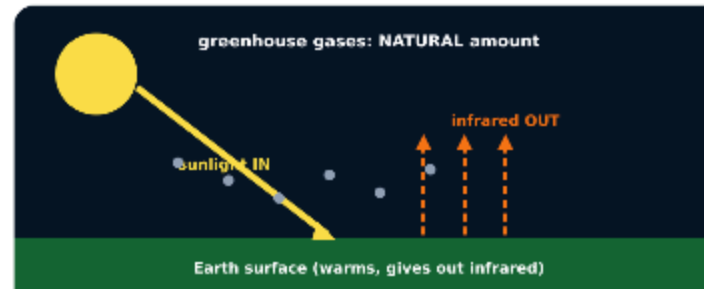
CLIMATE · a pattern over many years

ENERGY · in, out and re-emitted

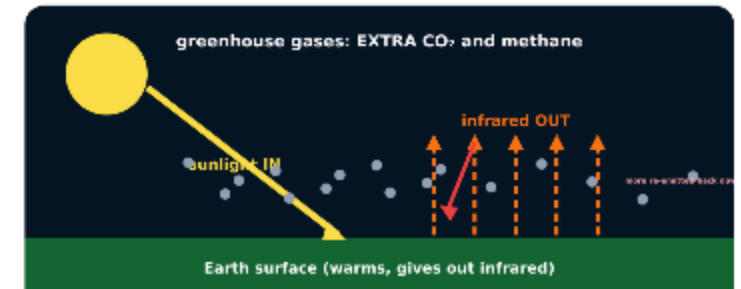
Look closely · what is the evidence?



Weather versus climate



Natural greenhouse effect



Enhanced greenhouse effect

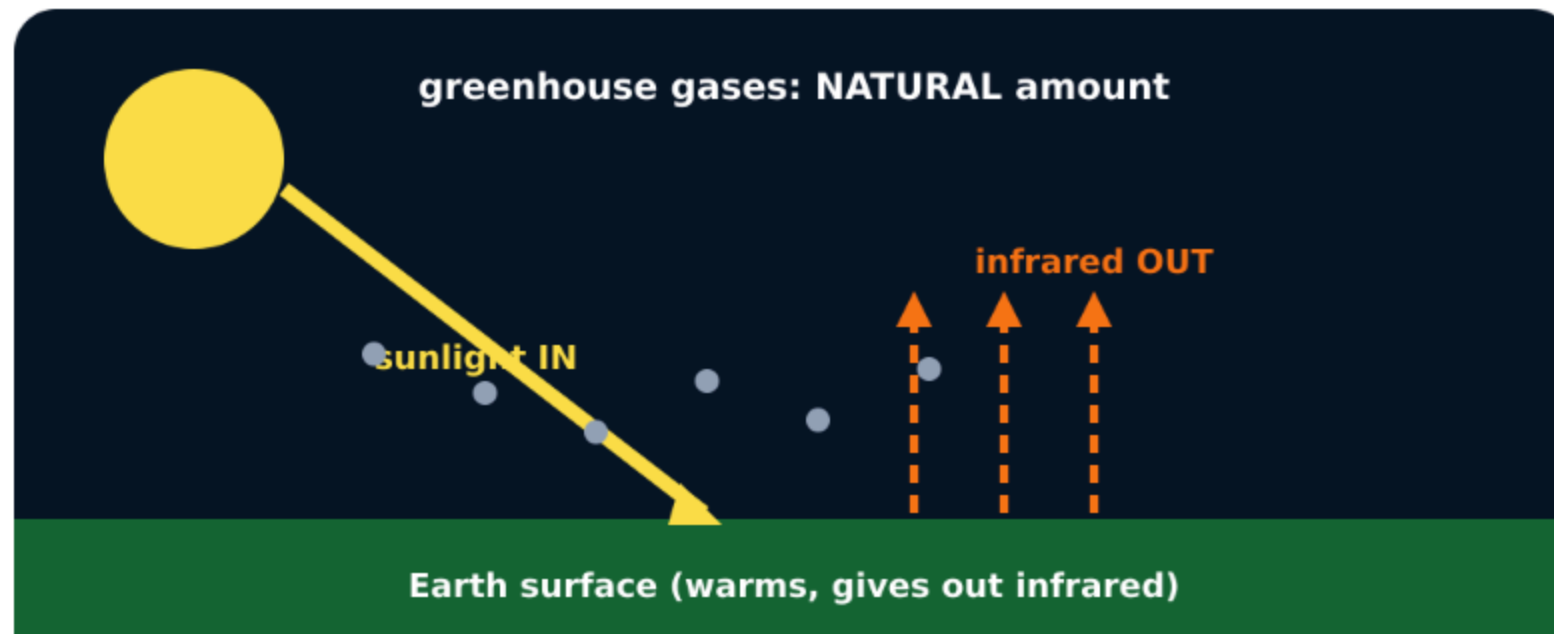
Words we will use

- climate — weather pattern over time
- greenhouse gas — gas that affects heat flow
- average — one value for a set

Watch the model

- Sunlight passes through the atmosphere and transfers energy to Earth's surface.
- The warm surface gives out infrared energy. Naturally occurring greenhouse gases absorb and re-emit some of it; this natural effect keeps Earth warm enough for life.
- Extra carbon dioxide and methane enhance the effect: more outgoing infrared is absorbed and some is re-emitted towards the surface, slowing net energy loss until the global average temperature rises.

Watch the model



Greenhouse effect model

Which statement is scientific?

- A. Climate is the usual pattern over many years; one cold day is weather.
 - B. Global warming means every place becomes hotter every day.
 - C. The ozone hole is the main cause of global warming.
- Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Human activities such as burning coal, oil and gas, plus deforestation, add greenhouse gases.
- A stronger greenhouse effect raises the global average; it does not make every day or place equally hot.
- This changes risks: heat extremes, rainfall patterns, melting land ice, sea-level rise and habitat stress.

Find and repair the misconception

One cold week disproves global warming.

A. Keep it: one week shows the whole climate.

B. Repair it: a short weather event cannot overturn a long-term global pattern.

C. Swap it: global warming means every day must be hot.

Climate Chain Lab

Order one genuine five-link chain from a human activity to one likely effect. Say why each link follows the one before it.

- Human activity
- More greenhouse gas
- Energy loss slows
- Average temperature rises
- One likely effect

Climate Chain Lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- 1 · HUMAN ACTIVITY
- 2 · MORE GREENHOUSE GAS
- 3 · ENERGY CHANGE
- 4 · GLOBAL WARMING
- 5 · LIKELY EFFECT

Read all five links with because and can lead to. Keep the effect likely, not guaranteed everywhere.

Your independent mission

Create the five-link chain activity → more greenhouse gas → slower net energy loss → higher global average → one likely effect, then explain it aloud, by pointing or in writing.

Record: One ordered causal chain, one explained effect and one honest model limitation.

Choose your support and challenge

- Supported: Order the five icon-backed links and complete: More greenhouse gas slows _____. The higher global average can lead to _____.
- Standard: Draw or order the full chain and explain one effect using because.
- Stretch: Contrast the natural and enhanced greenhouse effects, then explain why one weather event is weak evidence.

Show what you know

What changes in the energy route when extra greenhouse gas is added?

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.